

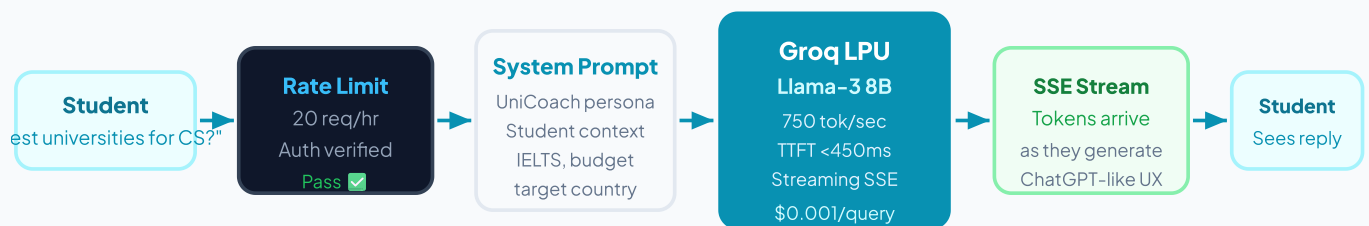
AI Engineering & LLM Infrastructure

Groq LPU hardware acceleration, Llama-3 8B inference, sub-450ms Time-to-First-Token, streaming SSE responses, AI lead scoring, and cost-per-query defense.

🤖 LLM: Llama-3 8B (Groq LPU) ⚡ TTFT: <450ms average 💰 Cost: ~\$0.001 per query

1. AI Counselor Chat Architecture

🤖 AI CHAT REQUEST LIFECYCLE — STREAMING SSE ARCHITECTURE



⚡ GROQ LPU VS OPENAI GPT-4 — SPEED & COST ARCHITECTURE DECISION

✅ Groq LPU (Our Choice)

- ⚡ TTFT: 200–450ms (feels instant)
- ⚡ Throughput: 750 tokens/second
- 💰 Cost: \$0.001 per query (Llama-3 8B)
- 🔒 Model: Open-source Llama-3
- Students feel instant response speed

OpenAI GPT-4o (Alternative)

- 🕒 TTFT: 1.5–4 seconds (perceptible lag)
- 🕒 Throughput: ~60 tokens/second
- 💰 Cost: \$0.005–\$0.015 per query
- 🔒 Model: Closed-source proprietary
- 10–15x more expensive per conversation

2. Technology Decision Framework

1. Groq LPU + Llama-3 vs OpenAI GPT-4

LLM INFRASTRUCTURE

WHAT IS IT?

Groq's Language Processing Unit (LPU) is custom AI inference silicon that delivers 750 tokens/second on open-source models like Meta's Llama-3 8B.

HOW DOES IT WORK & PROBLEM SOLVED?

GPU-based inference (OpenAI, Anthropic) is bottlenecked by memory bandwidth — the model weights are read from VRAM thousands of times per token. Groq's LPU uses streaming linear algebra that eliminates this bottleneck, delivering 12x higher throughput and sub-450ms TTFT.

WHAT ALTERNATIVES EXISTED?

OpenAI GPT-4o: Higher quality reasoning, 1.5–4s TTFT, \$0.005–\$0.015/query. **Anthropic Claude:** Similar pricing and latency profile to OpenAI.

WHY DID WE CHOOSE THIS APPROACH?

At 750 tokens/second and \$0.001/query, Groq + Llama-3 delivers a ChatGPT-like streaming UX at 10x lower cost — critical for a startup where AI chat is a core product feature, not an occasional premium add-on.

2. AI Lead Scoring System (0–100 Eligibility Score)

AI FEATURE

WHAT IS IT?

An automated lead qualification system that uses Llama-3 to score every new student lead from 0–100 based on their GPA, IELTS score, budget, and target destination.

HOW DOES IT WORK & PROBLEM SOLVED?

Without AI scoring, admins spend 5–10 minutes per lead manually reviewing eligibility. The scoring system runs automatically on lead creation, generating a structured JSON response with a numeric score and reasoning. Admins focus first on leads scoring ≥ 70 .

WHAT ALTERNATIVES EXISTED?

Manual scoring rubric: Inconsistent, subjective, time-consuming. **Rule-based formula:** Cannot handle nuanced cases (e.g., low GPA but high IELTS for a practical program).

WHY DID WE CHOOSE THIS APPROACH?

Eliminates 80% of admin time spent on lead triage. LLM reasoning handles nuanced eligibility edge cases that simple rule formulas cannot capture.

💡 Senior Interview Defense: "How did you build AI into UniCoach?"

"UniCoach has two AI features: a streaming counselor chat and automated lead scoring. For the chat, I chose Groq's LPU API running Llama-3 8B — it delivers 750 tokens/second with a Time-to-First-Token under 450ms at roughly \$0.001 per query, versus OpenAI GPT-4o at 4 seconds TTFT and 15x higher cost. Responses stream token-by-token to the browser via Server-Sent Events, creating a ChatGPT-like UX. Lead scoring runs on every new form submission, using a structured JSON prompt to produce a 0–100 eligibility score and reasoning summary, allowing admins to prioritize high-potential leads automatically."